

AA-THEO LA

Theophylline Anhydrous

Sustained-release Tablets 200 and 300mg

Bronchodilator

PHARMACOLOGY: Theophylline relaxes the smooth muscle of the bronchial airways and pulmonary blood vessels to relieve bronchospasm and increase flow rates and vital capacity. It also produces other actions typical of the xanthine derivatives: coronary vasodilatation, diuresis, increase in gastric secretion, and cardiac, cerebral, and skeletal muscle stimulation. The actions of theophylline may be mediated through inhibition of phosphodiesterase and a resultant increase in intracellular cyclic adenosine monophosphate, but the exact mechanism(s) has not been determined. Theophylline is usually readily absorbed and distributed into all body compartments. Protein binding accounts for some 55 - 65%. The liver is the primary site of metabolism. The therapeutic serum concentration of theophylline is accepted as 10 to 20µg/mL (55 - 110µmol/L); levels above 20µg/mL are associated with toxic reactions. The pharmacokinetics of theophylline are influenced by a number of variables such as: age, disease state, smoking, concomitant medication. Therefore the optimum therapeutic maintenance dose should be determined by individual titration.

AA-THEO LA Tablets are sustained-release tablets which produce peak blood levels between 5 - 8 hours after dosing in adults. Once the steady state level has been reached (3 days), the therapeutic blood levels persist for 12 hours in most adult patients. The mean elimination half-life of theophylline in nonsmoking adults is about 8 hours. The degree of fluctuation between peak trough theophylline levels can be defined as follows:

% Theophylline Fluctuation	Nonsmoking Adults
Theophylline sustained-release tablets 200, 300mg	27%
Theophylline sustained-release tablets 100mg	34%

INDICATIONS: AA-THEO LA (theophylline) tablets are indicated for the symptomatic treatment of reversible bronchospasm associated with asthma, chronic bronchitis, emphysema and related bronchospastic disorders.

ADVERSE REACTIONS: The most common adverse reactions are nausea, vomiting, epigastric pain, headache and tremor. These are usually early signs of toxicity; however, with high doses cardiac arrhythmias or seizures may be the first signs to appear. Adverse reactions reported with theophylline preparations include:

Gastrointestinal: nausea, vomiting, epigastric pain, hemoatemesis, diarrhea, anorexia, reactivation of peptic ulcer, intestinal bleeding.

Central Nervous System: headaches, irritability, restlessness, insomnia, hyperactivity, reflex hyperexcitability, muscle twitching, clonic and tonic generalized convulsions.

Cardiovascular: palpitation, tachycardia, extrasystoles, flushing, hypotension, circulatory failure, life-threatening ventricular arrhythmias.

Respiratory: tachypnea.

Renal: albuminuria, diuresis and hematuria.

Others: hyperglycemia and inappropriate ADH syndrome.

WARNINGS: In clinical situations where immediate bronchodilatation is required, such as status asthmaticus, AA-THEO LA (theophylline) is not suitable.

Since theophylline has a narrow therapeutic index, the margin of safety above therapeutic doses is small. In patients showing intolerance to theophylline, the therapy should be reassessed.

Theophylline clearance can be changed by various disease states, as well as by the age of the patient, concomitant use of other medications and lifestyle habits (see PRECAUTIONS).

The use of AA-THEO LA in children under the age of twelve years is not recommended.

PRECAUTIONS: AA-THEO LA (theophylline) tablets should be swallowed whole. Do not break, chew or crush.

Marked differences in serum levels may be seen in patients receiving the same theophylline dose. This may be explained by differences between patients in the rate of metabolism.

Smokers and children are usually high metabolizers. Dosage regimens should therefore be individualized.

Theophylline half-life is shorter in smokers than in nonsmokers. Smokers may require larger or more frequent doses of theophylline.

Ideally, serum theophylline levels should be monitored in all patients and

a theophylline half-life calculated which would enable doses and dosing regimens to be tailored to each patient to maintain a therapeutic level, to ensure optimal clinical response and to avoid toxicity.

The incidence of toxicity increases at serum theophylline levels greater than 15mg/mL (82.5µmol/L) and levels above 20µg/mL (110µmol/L) are usually quite toxic in most adult patients. High serum levels may be seen in some patients receiving doses considered to be conventional. The possibility of overdose should therefore not be considered with large doses only. Overdosage of theophylline may cause peripheral vascular collapse.

Reduced theophylline clearance has been documented in the following readily identifiable groups:

1. patients with impaired renal or hepatic function;
 2. patients over 55 years of age, particularly males and those with chronic lung disease;
 3. those with cardiac failure from any cause;
 4. patients taking certain drug (i.e. macrolide antibiotics and cimetidine).
- Decreased clearance may be associated with either influenza immunization or active infection with influenza.

Laboratory monitoring of serum theophylline is especially appropriate in the above individuals to maintain an appropriate theophylline dosage.

Serious side effects such as tachycardia, arrhythmia, seizures, vascular collapse and even death may occur without warning and may not be preceded by less severe symptoms such as nausea and restlessness.

Use with caution in patients with severe cardiac disease, severe hypoxemia, hypertension, hyperthyroidism, acute myocardial injury, cor pulmonale, congestive heart failure, liver disease, in the elderly (especially males).

Patients with congestive heart failure frequently have markedly prolonged serum levels with theophylline persisting in serum for long periods following discontinuation of the drug.

Theophylline may occasionally act as a local irritant to the gastrointestinal tract although gastrointestinal symptoms are more common centrally mediated and associated with serum drug concentrations over 20µg/mL (110µmol/L).

Theophylline increases gastric secretion, and caution should be exercised in patients with a history of peptic ulcer.

Although AA-THEO LA has pharmacokinetic properties similar to other controlled-release theophylline products, it is not possible to ensure interchangeability between different products. Careful clinical monitoring is required when changing from one drug product to another.

The concurrent administration of other theophylline derivatives along with AA-THEO LA is not recommended.

Laboratory Test Interactions

In the interpretation of biochemical tests, it should be remembered that theophylline may cause an elevation of urine catecholamines and plasma free fatty acids.

When plasma levels of theophylline are measured by spectrophotometric methods, coffee, tea, cola beverages, chocolate and acetaminophen contribute to falsely high values.

When a high pressure liquid chromatography (HPLC) method is used, plasma theophylline concentration may be falsely increased by caffeine, some cephalosporin and sulfa medications.

Food Interaction

Theophylline clearance is increased when diet includes a low carbohydrate, high protein intake, or a high carbohydrate, low protein intake and there is a chronic ingestion of charcoal broiled meats. However, the administration of AA-THEO LA with meals appears not to significantly effect the amount of theophylline released from AA-THEO LA tablets.

Usage in Pregnancy and Lactation

Theophylline crosses the placental barrier and also passes freely into breast milk, where concentrations are similar to plasma levels. Safe use in pregnancy has not been established relative to possible adverse effects on fetal development, but neither have adverse effects on fetal development been established. Therefore, use of theophylline for uncontrolled asthma in pregnant women and nursing mothers should be balanced against the risk of potential effects on the fetus or on the nursing newborn.

CONTRAINDICATIONS: AA-THEO LA (theophylline) is contraindicated in patients with:

- hypersensitivity to theophylline or xanthine derivatives;
- peptic ulcer;
- coronary artery disease (when, in the physicians judgement, myocardial stimulation might prove harmful).

DRUG INTERACTIONS:

A theophylline pharmacokinetics are altered by the concurrent use of various drugs as listed below:

DRUG	THEOPHYLLINE
a) cimetidine, propranolol, allopurinol, macrolide antibiotics (erythromycin), oral contraceptives.	↑ t½, ↓ clearance
b) Alkalinizing agents	↑ t½, ↓ clearance
c) Influenza vaccine	↑ t½, clearance reported to be decreased or no change.
d) Phenytoin, barbiturates, carbamazepine, isoproterenol, rifampin	↓ t½, ↑ clearance
e) Smoking (tobacco)	↓ t½, ↑ clearance
f) Acidifying agents	↓ t½, ↑ clearance

B. Concurrent use of theophylline influences effects of certain drugs :

DRUG	THEOPHYLLINE
a) Digitalis glycosides	↑ cardiac effect
b) Thiazides	↑ diuresis
c) Nephrotoxic drugs	↑ nephrotoxicity
d) Lithium	↑ ratio of lithium / creatinine clearance, thus decrease serum lithium
e) Sympathomimetic amines	↑ toxicity, ↑ CNS stimulation
f) Coumarin anticoagulants	↓ anticoagulant activity, increase prothrombin and fibrinogen blood concentrations, shorten prothrombin time
g) Allopurinol	↓ antihyperuricemic action
h) Probenicid and pyrazolon derivatives	↓ uricosuric action
i) Ketamine	↓ threshold value for inducing convulsions

DOSEAGE AND ADMINISTRATION: Therapeutic serum levels are generally considered to be between 10 and 20µg/mL (55µmol/L and 110µmol/L). Due to variable rates of elimination, there is patient-to-patient variation in dosage needed to achieve a therapeutic serum level. Because of the variation from patient to patient, the variation within the same patient, and the relatively narrow therapeutic range, dosage should be individualized. Monitoring of serum theophylline concentrations is also extremely important, especially in the initial stages of therapy (see PRECAUTIONS).

It is preferable to monitor peak concentrations rather than trough concentrations. Therefore, blood samples should be drawn 4 - 8 hours after AA-THEO LA (theophylline) dosing. It should be ascertained within 3 days). Depending on the sensitivity of the assay method used, dietary xanthines may interfere with assay results. If a dosage increase is not tolerated, dosage should be reduced to the previously tolerated level. Do not attempt to maintain dosage which is not tolerated or which produces serum concentrations above the therapeutic range.

AA-THEO LA tablets should be swallowed whole. Do not break, chew or crush.

Adults

The usual initial adult dose is 200 - 300mg every 12 hours. This dose may be increased by 50 - 100mg every 12 hours at 3 day intervals until a satisfactory response is obtained or toxic effects appear.

Dosage adjustments should be based upon serum theophylline concentration and / or upon the patient's clinical response. However, doses of 450mg every 12 hours or higher (900mg / day) should not be given unless serum theophylline concentration can be monitored. It should not be necessary to exceed a daily dose of 16mg/kg in adult patients. Even with serum level monitoring, this dose may lead to side effects because of day-to-day variations in blood levels within individual patients.

Children

The use of AA-THEO LA in children under the age of twelve years is not recommended.

SYMPTOMS AND TREATMENT OF OVERDOSAGE:

Symptoms: Insomnia, restlessness, mild excitement or irritability, and rapid pulse, are early symptoms which may progress to mild delirium. Sensory

disturbances such as tinnitus or flashes of light are common. Anorexia, nausea and vomiting are frequently early observations of theophylline overdose.

Fever, diuresis, dehydration and extreme thirst may be seen. Severe poisoning results in bloody, syrup-like 'coffee-ground' vomitus, tremors, tonic extensor spasm interrupted by clonic convulsions, extrasystoles, quickened respiration, stupor and finally coma.

Cardiovascular disorders and respiratory collapse, leading to shock, cyanosis and death follow gross overdosages.

Treatment:

A. Monitoring serum theophylline levels

Following intake of theophylline sustained-release tablets, the blood theophylline peak levels may not show until 5 - 8 hours post-ingestion in adults. Patients ingesting overdoses of sustained-release theophylline formulations may have, after the initial rise in blood theophylline, also a secondary increase in theophylline levels. One report on fatal self-poisoning has attributed this to compacted tablet masses in the gastrointestinal tract. Careful clinical and laboratory monitoring of stabilized patients is advisable.

B. If potential oral overdose is established and seizure has not occurred:

- 1) induce vomiting;
- 2) administer a cathartic (this is particularly important when a sustained-release preparation has been taken);
- 3) administer activated charcoal.

C. If the patient is having a seizure:

- 1) establish an airway;
- 2) administer oxygen;
- 3) treat the seizure with intravenous diazepam, 0.1 to 0.3mg/kg up to a total dose of 10mg;
- 4) monitor vital signs, maintain blood pressure and provide adequate hydration.

D. Post-seizure coma:

- 1) maintain airway and oxygenation;
- 2) if a result of oral medication, follow above recommendations to prevent absorption of drug, but intubation and lavage will have to be performed instead of inducing vomiting, and the cathartic and charcoal will need to be introduced via a large bore gastric lavage tube;
- 3) continue to provide full supportive care and adequate hydration while waiting for drug to be metabolized. In general, the drug is metabolized sufficiently rapidly so as not to warrant consideration of dialysis. However, if serum levels exceed 50µg/mL (257µmol/L), charcoal hemoperfusion may be indicated

Mode of Administration: Oral

AVAILABILITY OF DOSAGE FORMS: AA-THEO LA formulated as sustained-release tablets contains anhydrous theophylline with no colour additives. AA-THEO LA is available in two strengths: 200mg and 300mg.

AA-THEO LA 200mg: White, oval, biconvex, tablets, scored and engraved "THE 200" on one side, and "APO" on the other. Available in HDPE bottles of 60 tablets and 100 tablets.

AA-THEO LA 300mg: White, capsule-shaped, biconvex, tablets, scored and engraved "THE 300" on one side, and "APO" on the other. Available in HDPE bottles of 60 tablets and 100 tablets.

STORAGE: Store below 30°C

Manufacturer: Apotex Inc., 150 Signet Drive, Weston, Ontario, Canada M9L 1T9.

Date of Revision of Package Insert: June 2021



Apotex Inc.

150 Signet Drive, Weston
Ontario, Canada M9L 1T9

Distributor : Pharmaforte (M) Sdn Bhd